

Isaac Center for Public Health

Indian Institute of Science, Bengaluru

Platform Product Requirements Document

Integrated Public Health Data and Geospatial Platform

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Geospatial Data Portal · Cohort Study Management · Sample Tracking · Reproducible Analysis Pipelines



01 Executive Summary

The Isaac Center for Public Health requires a unified digital platform to support two (at first) distinct but complementary research and operational capabilities: (1) an integrated geospatial and public health data portal for location-aware analysis across India, and (2) a longitudinal cohort study management system for the ICPH Pregnancy Cohort. This document defines the product requirements, functional specifications, and non-functional expectations for both modules as a single deployable platform.

Module A — the Geospatial & Public Health Data Portal — enables analysts, researchers, and programme staff to discover, summarise, visualise, and merge large publicly available geospatial datasets (satellite, meteorological, demographic, land-cover) with health datasets reported at administrative spatial units or carrying point geolocations. It covers the geographic extent of India for 2000–2025.

Module B — the Cohort Study Management System — provides end-to-end operational support for the ICPH Pregnancy Cohort: mobile-capable offline data collection, longitudinal data management with plausibility checks, study dashboard with visit tracking, participant reminders, and a sample tracking system from collection through to long-term storage at IISc.

02 Product Vision & Goals

The platform vision is to make rigorous, reproducible, location-aware public health research accessible to researchers and programme staff without requiring bespoke data engineering for each task, while simultaneously providing robust operational infrastructure for running long-term cohort studies at Isaac.

Strategic Goals

- Lower the technical barrier to working with heterogeneous geospatial and health data.
- Enable reproducible, auditable analysis pipelines from raw data to publication-ready outputs.
- Provide a single point of access to curated, harmonised geospatial datasets for India (2000–2025).
- Support the full operational lifecycle of longitudinal cohort studies from recruitment through sample archive.
- Establish interoperable, standards-based data services extensible to future studies and datasets.

03 Stakeholders & User Roles

Role	Responsibilities & Access
Public / Guest	Browse the catalogue; view selected public summaries and maps (read-only, no registration required).
Analyst / Researcher	Run spatial and temporal summarisations; build maps and tables; register, call, and merge health datasets; export outputs. Full Module A access.
Study Coordinator / Nurse	Access the cohort study dashboard; view due-visit lists; record study events; manage participant reminders (Module B).
Data Collector (Field Staff)	Submit ODK-style forms via mobile device, online or offline; view own submission queue.
Data Steward	Curate geospatial datasets; manage metadata and the administrative boundary crosswalk; manage health dataset registry.
Principal Investigator	Access all study-level dashboards and summary reports; review flagged data quality issues.
Platform Administrator	Manage users and roles; configure governance policies; monitor system health and configuration.

Table 1: Platform User Roles

04 Platform Architecture Overview

The platform is composed of shared infrastructure serving both modules, with module-specific service layers exposed through a unified web application. Technology choices for each layer will be confirmed during detailed design.

Layer	Description
Ingestion & Processing	Automated pipelines to acquire source geospatial data, clip to India, harmonise CRS, and generate temporal aggregations and metadata. Separate pipelines for cohort form submissions and lab/sample uploads.
Storage	Scalable object/array storage for cloud-optimised rasters and time series; relational/spatial database for boundaries, metadata catalogue, registered health datasets, cohort records, and sample tracking logs.
Services & APIs	OGC-compliant map and feature services; summarisation and spatial-join APIs; health dataset call/merge services; cohort data and study-management APIs; authentication and audit services.

Layer	Description
Application (Web Portal)	Catalogue search; interactive maps; tabular data explorer; merge configuration tools; cohort study dashboard; offline-capable data collection interface; sample tracking interface.
Governance & Security	Role-based authentication and authorisation; audit logging for all data access and mutation; data governance controls for health datasets; encryption in transit and at rest.

Table 2: Platform Architecture Layers

MODULE A

Geospatial & Public Health Data Portal

Discovery, summarisation, visualisation, and integration of geospatial and public health datasets for India (2000–2025).

05 Data Sources & Repository

The portal's core repository holds curated, analysis-ready copies of the following publicly available geospatial data sources, harmonised to a common spatial reference (geographic CRS) and clipped to the Indian national boundary for 2000–2025. Each source is documented with provenance, native resolution, temporal coverage, and processing lineage.

Data Source	Category	Key Variables	Resolution	Coverage
LULC	Land Cover	Cropland, built-up, forest, water, barren; class change	10–500 m	2000–2025 (annual)
TERRA (MODIS)	Satellite	Reflectance, NDVI/EVI, LST, aerosol depth, land cover	250 m – 1 km	2000–2025 (daily–16 day)
AQUA (MODIS)	Satellite	Reflectance, NDVI/EVI, LST, aerosol optical depth	250 m – 1 km	2002–2025 (daily–16 day)
Sentinel	Satellite	S2 optical (reflectance, NDVI); S5P trace gases	10–60 m / ~7 km	2015–2025 (5–10 day)
ERA5 Reanalysis	Meteorological	Temperature, precipitation, humidity, wind, radiation	~0.25° (~31 km)	2000–2025 (hourly/daily)

Data Source	Category	Key Variables	Resolution	Coverage
OpenStreetMap	Vector	Roads, buildings, POIs, admin & natural features	Vector	2000–2025 (continuous)
Gridded Population	Demographic	Population count, density, age/sex structure	~100 m – 1 km	2000–2025 (annual)
CHAIR-India PM2.5 & Temp	Air Quality / Climate	Surface PM2.5 concentration; near-surface temperature	1 km × 1 km	2008–2020 (daily)

Table 3: Curated Geospatial Data Sources

Repository Requirements

- Store all datasets in a common documented spatial CRS with consistent national extent for India.
- Preserve native temporal resolution and provide standardised temporal aggregations (daily, monthly, seasonal, annual).
- Maintain a metadata catalogue recording source, provenance, units, resolution, coverage, and processing lineage.
- Support cloud-optimised, scalable storage formats (e.g., COG, Zarr) for large raster and time-series data.
- Version datasets so updates and reprocessing are tracked and reproducible.

06 Module A – Functional Requirements

FR-A-01 Data Discovery & Catalogue

ID	Requirement	Priority
FR-A-D IS-01	Browse and search the dataset catalogue by source, theme, spatial unit, and time period.	Must Have
FR-A-D IS-02	View metadata and documentation for each dataset, including provenance, units, and processing lineage.	Must Have
FR-A-D IS-03	Filter the catalogue by geographic area of interest: national, state, district, subdistrict, ward, or user-supplied boundary.	Must Have
FR-A-D IS-04	Display dataset versioning history and flag when a dataset has been reprocessed since a user's last access.	Should Have

FR-A-02 Summarisation of Variables

ID	Requirement	Priority
FR-A-S UM-01	Compute spatial summaries (mean, min, max, sum, percentiles, area-weighted statistics) of any variable over a selected area or spatial unit.	Must Have
FR-A-S UM-02	Compute temporal summaries: monthly means, seasonal aggregates, annual trends, anomalies relative to a configurable baseline.	Must Have
FR-A-S UM-03	Perform zonal statistics aggregating raster variables to administrative units (district, subdistrict, ward) or user-supplied polygons.	Must Have
FR-A-S UM-04	Allow users to define, name, and save custom summarisation parameter sets for reuse across sessions.	Should Have
FR-A-S UM-05	Run heavy summarisation jobs asynchronously with progress feedback and email/in-app notification on completion.	Must Have
FR-A-S UM-06	Record every summarisation operation with all parameters and data versions to support reproducibility.	Must Have

FR-A-03 Map Visualisation

ID	Requirement	Priority
FR-A-M AP-01	Display selected variables as interactive map layers with pan, zoom, and area-of-interest selection.	Must Have
FR-A-M AP-02	Render raster variables (satellite indices, temperature, PM2.5) and choropleth maps of summarised values at district, subdistrict, or ward level.	Must Have
FR-A-M AP-03	Provide configurable symbology: colour ramps, class breaks, opacity control, legend, scale bar, and selectable basemaps.	Must Have
FR-A-M AP-04	Provide a time slider or animation control to view a variable across the 2000–2025 period.	Must Have
FR-A-M AP-05	Overlay multiple layers (e.g., a health indicator over a meteorological variable) with adjustable per-layer transparency.	Must Have
FR-A-M AP-06	Export the current map view as an image (PNG/PDF) and underlying summarised values as downloadable data.	Must Have
FR-A-M AP-07	Expose OGC-compliant map services (WMS/WMTS) for integration with external GIS tools.	Should Have

FR-A-04 Tabular Display & Export

ID	Requirement	Priority
FR-A-T AB-01	Present summarised variables in sortable, filterable tables keyed by spatial unit and time period.	Must Have
FR-A-T AB-02	Support pivoting between spatial units and time periods (e.g., districts as rows, years as columns).	Must Have
FR-A-T AB-03	Display units, source, and the summarisation method applied for each column.	Must Have
FR-A-T AB-04	Export tables to CSV and Excel with an auto-generated data dictionary in the export package.	Must Have

FR-A-05 Health Dataset Integration — Call & Merge

Two classes of health data are supported: (a) datasets reported at administrative spatial units (district, subdistrict, ward), and (b) datasets carrying point geolocations (latitude/longitude).

ID	Requirement	Priority
FR-A-M RG-01	Allow users to register health datasets from supported repositories or direct file uploads (CSV, Excel, spatial formats) with documented schema and spatial key fields.	Must Have
FR-A-M RG-02	For spatial-unit datasets, merge on standardised administrative codes or names (district, subdistrict, ward) with validation against the authoritative boundary reference.	Must Have
FR-A-M RG-03	For geolocated datasets, perform spatial joins assigning each point to its containing administrative unit and/or extracting geospatial variable values at the point location.	Must Have
FR-A-M RG-04	Produce a combined analysis table from selected health indicators and geospatial variable summaries for a common area and time period.	Must Have
FR-A-M RG-05	Flag and report unmatched records, mismatched units, and CRS issues during merge, with a review-and-resolve UI.	Must Have
FR-A-M RG-06	Save and reuse merge configurations across sessions, and record full lineage of every merged output.	Must Have
FR-A-M RG-07	Support batch merge operations for merging multiple health datasets against a shared geospatial summary table.	Could Have

FR-A-06 Spatial Unit Handling

ID	Requirement	Priority
FR-A-S PA-01	Maintain an authoritative reference of Indian administrative boundaries (state, district, subdistrict, ward) with stable identifiers and boundary-change history for 2000–2025.	Must Have
FR-A-S PA-02	Provide a managed crosswalk to reconcile differing naming conventions and code systems across data sources.	Must Have
FR-A-S PA-03	Allow users to upload custom boundary polygons for area-of-interest analysis.	Should Have
FR-A-S PA-04	Expose authoritative boundary layers as OGC WFS endpoints for external use.	Could Have

MODULE B

Cohort Study Management System

ICPH Pregnancy Cohort — operational infrastructure for longitudinal study management, data collection, and sample tracking.

07 Study Context — ICPH Pregnancy Cohort

The ICPH Pregnancy Cohort is a longitudinal follow-up study of pregnant women recruited before 20 weeks gestational age (GA) at two hospitals in Chennai, followed until 6 months postpartum. The study involves mandatory visits at defined milestones, opportunistic contacts between mandatory visits, and the collection of both structured data and biological samples at each timepoint.

Mandatory Study Visit Schedule

Visit	Timing / Trigger	Data & Sample Collection
Baseline	Initial contact, < 20 weeks GA	Baseline data forms, blood samples
Visit 2	24–28 weeks GA	Study forms, blood samples (e.g., insulin tube)
Delivery Visit	At delivery	Delivery data, samples

Visit	Timing / Trigger	Data & Sample Collection
1 Month Postpartum	~1 month after delivery	Postpartum data, infant measurements, samples
6 Months Postpartum	~6 months after delivery — study end	Final data, infant measurements, samples

Table 4: Mandatory Study Visit Schedule

Between mandatory visits, study staff aim to collect diet, physical activity, and recent diagnosis data whenever participants attend the hospital for routine antenatal care.

08 Module B – Functional Requirements

FR-B-01 Data Collection

ID	Requirement	Priority
FR-B-C OL-01	Provide ODK-compatible electronic data capture forms, deployable on Android and iOS devices, supporting both online and fully offline operation with automatic upload on reconnection.	Must Have
FR-B-C OL-02	Support saving partially completed forms, with ability to resume at any point before final submission.	Must Have
FR-B-C OL-03	Support different form schemas per visit type (baseline, antenatal contact, 24–28 wk, delivery, 1 month postpartum, 6 month postpartum), configurable by data stewards.	Must Have
FR-B-C OL-04	Enforce field-level validation rules (required fields, numeric range checks, date logic) at point of entry, surfacing errors before submission.	Must Have
FR-B-C OL-05	Enable longitudinal plausibility checks: compare a new entry against the same participant's prior values and flag implausible changes in real time (e.g., infant length must be non-decreasing across visits), when internet is available.	Must Have
FR-B-C OL-06	Transmit completed and partially completed forms to the IISc server via encrypted connection; maintain a local submission queue if the server is unreachable.	Must Have
FR-B-C OL-07	Associate every form submission with the correct participant ID, visit type, and data collector identity.	Must Have

FR-B-02 Data Management

ID	Requirement	Priority
FR-B-D AT-01	Store all participant data in a central database on the IISc server, with automatic separation of identifiable and de-identified / linked-anonymised data layers.	Must Have
FR-B-D AT-02	Allow authorised users to download selected variables for a defined participant subset and time range for analysis.	Must Have
FR-B-D AT-03	Support upload and storage of lab result files (CSV and related tabular formats) from the laboratory service provider, linked to participant by study ID.	Must Have
FR-B-D AT-04	Support upload and storage of ultrasound images in DICOM, PDF, and PNG formats, linked to participant by study ID and visit date.	Must Have
FR-B-D AT-05	Run automated backend plausibility checks on time-dependent variables after each new submission, flag implausible values, and present flags in a review queue for data managers.	Must Have
FR-B-D AT-06	Maintain a full audit log of all data insertions, edits, and exports with timestamp and user identity.	Must Have
FR-B-D AT-07	Support export of de-identified analysis datasets in standard formats (CSV, Stata, SPSS, R-compatible).	Should Have

FR-B-03 Study Management & Dashboard

ID	Requirement	Priority
FR-B-M GT-01	Provide a study management dashboard displaying current counts of participants screened, enrolled, withdrawn, and completed – overall and stratified by hospital.	Must Have
FR-B-M GT-02	Display counts of participants due for each mandatory visit within a user-defined upcoming window, and counts who have completed each visit – overall and by hospital, with optional grouping by data collector.	Must Have
FR-B-M GT-03	Support both weekly and cumulative views of all summary statistics on the dashboard.	Must Have
FR-B-M GT-04	Generate a daily or weekly list of participants due for mandatory visits, taking participant withdrawal status into account (withdrawn participants must not appear in due-visit lists or generate reminders).	Must Have
FR-B-M GT-05	Provide configurable automated reminders to study coordinators and/or participants when a mandatory visit is due,	Must Have

ID	Requirement	Priority
	based on participant-specific visit windows derived from baseline data (e.g., estimated delivery date, actual delivery date).	
FR-B-M GT-06	Allow coordinators and nurses to access the dashboard and due-visit lists from a mobile-accessible interface.	Must Have
FR-B-M GT-07	Provide a participant-level timeline view showing completed, due, missed, and upcoming visits at a glance.	Should Have
FR-B-M GT-08	Support dashboard export as PDF or Excel for weekly study meetings.	Could Have

FR-B-04 Sample Tracking System

A blood sample tracking system is required for samples intended for long-term storage at IISc. The system must track each individual sample tube from collection through all intermediate steps to final storage.

ID	Requirement	Priority
FR-B-S AM-01	Register each sample tube with a unique tube ID linked to participant study ID, visit type, and collection date.	Must Have
FR-B-S AM-02	Record and track the following status events per tube: (1) Sample collected; (2) Sample sent to lab; (3) Sample analysed for immediate tests (e.g., insulin, glucose); (4) Remaining sample aliquoted at lab with aliquot tube IDs recorded; (5) Sample dispatched to NIE for storage; (6) Sample received at NIE; (7) Sample dispatched to IISc for long-term storage; (8) Sample received at IISc.	Must Have
FR-B-S AM-03	Record the sample type/assay context per tube (e.g., 'baseline basal OGTT tube', '24–28 wk insulin tube') using a configurable controlled vocabulary.	Must Have
FR-B-S AM-04	Record aliquot tube IDs (child tubes) linked to parent tube ID for samples that have been split at the lab.	Must Have
FR-B-S AM-05	Provide a sample tracking dashboard showing counts of tubes at each status stage, filterable by participant, visit, sample type, and date range.	Must Have
FR-B-S AM-06	Flag samples that are overdue at any step after a configurable threshold period.	Should Have
FR-B-S AM-07	Support barcode or QR-code scanning for tube ID entry via mobile device to reduce manual transcription errors.	Should Have

ID	Requirement	Priority
FR-B-S AM-08	Export complete sample manifests and chain-of-custody logs as CSV or Excel for audit purposes.	Must Have

09 Non-Functional Requirements

Category	Requirement
Performance	Interactive map and table queries on pre-summarised data return within 5 seconds for typical single-state areas. Heavy jobs run asynchronously. Data collection forms open and submit within 3 seconds on a standard mobile connection.
Scalability	Storage and compute scale to multi-decade, national-extent raster and time-series data (tens of TBs). Cohort database supports up to 10,000 participants with all associated samples, images, and lab results.
Reliability & Availability	Target $\geq 99.5\%$ uptime for the web portal and APIs during business hours. Automated daily backups with point-in-time recovery. Defined RTO and RPO documented in operations runbook.
Security	Role-based access control (RBAC) for all platform functions. TLS 1.2+ for all data in transit. AES-256 encryption at rest for health and sample records. MFA for all non-guest roles.
Data Governance & Privacy	Health datasets stored and accessed under defined governance approvals. PII stored separately from de-identified research data, restricted to authorised roles. De-identification applied before any analysis export.
Interoperability	OGC-compliant WMS/WMTS/WFS endpoints for geospatial data. Documented REST APIs for all programmatic access. ODK XForms-compatible data collection. Export in open formats (CSV, GeoPackage, GeoTIFF).
Reproducibility	Every summarisation, merge, and analysis operation logged with input dataset versions, parameters, and execution timestamp. Users can re-run or inspect any prior operation.
Usability	Interface usable by non-GIS specialists. Key workflows completable without reference to documentation after a one-hour onboarding. WCAG 2.1 AA accessibility compliance for the web portal.
Offline Capability	Data collection forms must function fully offline for up to 7 days without any connectivity, with reliable sync on reconnection and no data loss.

Table 5: Non-Functional Requirements

10 Integration Points Between Modules

The two modules share infrastructure but are designed to operate independently. The following integration points should be considered in the design phase.

- Cohort participants carrying geolocations can be fed into Module A's spatial join capability to extract local environmental exposures (PM2.5, temperature, NDVI) for each participant at each visit, enabling direct environment-health analysis within the platform.
- Summarised geospatial variables at the district or ward level of the two Chennai hospitals can be automatically appended to cohort data exports, allowing researchers to include contextual environmental variables without separate data wrangling.
- A shared user management and authentication layer applies consistently across both modules — a single login grants access to all permitted functions in Module A and Module B.
- A shared audit logging infrastructure captures all operations across both modules in a unified audit trail accessible to administrators.

11 Assumptions & Constraints

- Source geospatial datasets remain publicly available under licences permitting redistribution of derived summaries. If a licence is revoked, the affected dataset will be removed.
- Geospatial coverage is limited to India for 2000–2025; some sources (e.g., Sentinel) commence partway through this window. Temporal gaps are documented in metadata.
- Native resolutions differ across sources; harmonisation to common spatial units introduces documented approximations.
- Health datasets registered by users must carry adequate governance approvals, data quality documentation, and spatial keys appropriate for merging.
- Administrative boundaries change over time; analyses spanning boundary changes rely on the maintained crosswalk. Users are responsible for interpreting results in the context of known changes.
- Cohort study collection hardware (mobile devices) is the responsibility of the study team. The platform provides software infrastructure only.
- The IISc server environment meets the minimum compute and storage specifications to be determined during technical design.

12 Out of Scope

- Primary collection of new field or remote-sensing survey data.
- Clinical decision-making support or individual patient-level diagnosis tools.
- Hosting of personally identifiable health records beyond de-identified, spatially aggregated, or appropriately governed datasets.
- Geospatial data for countries outside India.

- Statistical modelling or analysis beyond spatial and temporal summarisation (users export data for analysis in R, Stata, Python, etc.).
- Real-time streaming data ingestion (all data is batch-ingested on defined schedules).

13 Open Questions – Items for Design Phase

#	Open Question
OQ-01	What are the precise server specifications and deployment environment at IISc? (Cloud, on-premise, hybrid?)
OQ-02	Which ODK platform variant will be used – ODK Central, KoboToolbox, or a custom ODK-compatible server – given the requirement to host on IISc infrastructure?
OQ-03	Should the NIE sample dispatch step (FR-B-SAM-02, steps 5–6) be included in v1 or deferred – to be confirmed by PI.
OQ-04	What are the governance and ethics board requirements for data storage location and cross-border data transfer (if any cloud provider outside India is used)?
OQ-05	What is the preferred notification channel for participant visit reminders – SMS, WhatsApp, email, or in-app? Any constraints from IRB/ethics approval?
OQ-06	What is the expected cohort size and recruitment rate, to properly size the cohort database and notification system?
OQ-07	Is barcode printing infrastructure (label printers) available at collection sites for tube-level sample tracking (FR-B-SAM-07)?
OQ-08	Which health dataset repositories should be supported for the 'call' operation in FR-A-MRG-01 (e.g., HMIS, SRS, NFHS, custom APIs)?

Table 6: Open Questions for Design Phase

14 Revision History

Version	Date	Author	Changes
0.1 (Draft)	10 June 2026	Siddhartha Mandal	Initial requirements capture from faculty input documents

Version	Date	Author	Changes
1.0 (Draft)	10 June 2026	—	Consolidated PRD with structured functional requirements, module architecture, and open questions

Table 7: Document Revision History